MIL-M-16034/2B
20 October 1964
SUPERSEDING
MIL-M-16034/002A(SHIPS)
16 January 1964
MIL-M-16034/2
2 January 1953

MILITARY SPECIFICATION

AMMETERS, D. C. SWITCHBOARD, 4-1/2 INCH, 250-DEGREE SCALE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for 250-degree scale, 4-1/2-inch direct current (d.c.) switchboard ammeters.

2. APPLICABLE DOCUMENTS

2.1 The following specification, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification, to the extent specified herein:

SPECIFICATION

MILITARY
MIL-M-16034 - Meters, Electrical Indicating, Switchboard and Portable Type.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements.- The general requirements shall be in accordance with MIL-M-16034, and as specified herein.

3.2 Design, construction, and physical dimensions.- The design, construction, and physical dimensions of the ammeters shall be as shown on figure 1.

3.3 Ranges.- The ammeters shall be furnished in the ranges shown in table I.

Table I - Ranges.

	Milliamperes, d.c.			Amperes, d.c.[1/]			Kiloamperes d.c.[1/]	Microamperes d.c.
Full-scale ranges	1	10	100	5	50	300	1	100
	2	20	200	10	80	500	1.5	200
	5	50	500	20	100	600	2	500
				30	800	800	3	50

[1/] Ammeters above 50 amperes and all kiloammeters are calibrated for use with 50-millivolt external shunts.

FSC 6625

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3.4 Detail requirements.- The detail requirements shall be as shown in table II.

Table II - Detail requirements.

Item	Requirement[1/]
Scale length	6.5 inches min.
Position influence (balance)	1 percent of scale length, max. change
Initial accuracy	1 percent max. error (1-1/2 percent max. error for high-impact-shock-resistant meters)
Damping factor	5 min.
Response time	2 seconds max.
Power consumption	500 mv. max.: milliammeters 150 mv. max.: self-contained ammeters 50 mv. ± 1 percent, external-shunt type ammeter (meter with 0.065 ± .010-ohm leads attached)
External-field influence	1 percent max. change in 5-oersted field
Heat effect at 65°C.	4 percent max. change at 65°C. .5 percent max. permanent change
Temperature influence	1 percent max. change
Low-temperature exposure	2.5 percent max. change at 0°C. .5 percent max. permanent change
Momentary overload	1 percent max. change
Sustained overload	1 percent of scale length, max. temporary zero shift .5 percent of scale length, max. permanent zero shift .33 percent max. permanent change in indication
Vibration	1 percent max. change
Friction effect	2 percent max. change
Humidity resistance	2 percent max. change
Shock (High-impact)	5 percent max. change

[1/] Percentages are based on full-scale value, unless otherwise indicated herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection.- Sampling and inspection shall be in accordance with MIL-M-16034.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-M-16034.

6. NOTES

6.1 Notes.- The notes specified in MIL-M-16034 are applicable to this specification.

Preparing activity:
Navy - SH
(Project 6625-0224)

Custodians:
Army - EL
Navy - SH
Air Force - 67

Review activities:
Army - EL, MU
Navy - SH
Air Force - 67, 11

User activities:
Army - MO, GL
Navy - MC
Air Force - None

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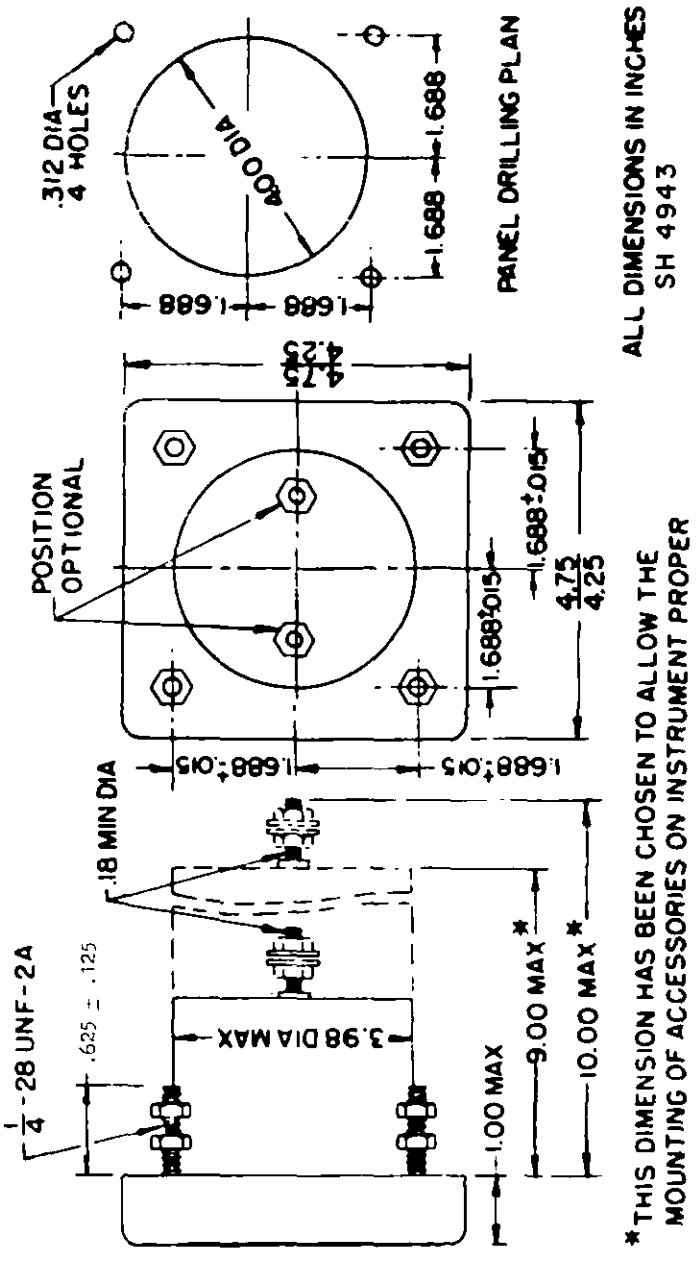


Figure 1 - Design, construction, and physical dimensions.

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